

Wizart DCC Platform: extensible USD-based toolset

Alexander Kalyuzhnyy
Wizart Animation
Russia
ak@wizartanimation.com

Konstantin Giliarovsky
Wizart Animation
Russia
kgiliarovsky@wizartanimation.com

Maksim Tokarev
Wizart Animation
Russia
mtokarev@wizartanimation.com

Mikhail Kutsov
Wizart Animation
Russia
mkutsov@wizartanimation.com

Vlad Tagintsev
Wizart Animation
Russia
vtagintsev@wizartanimation.com

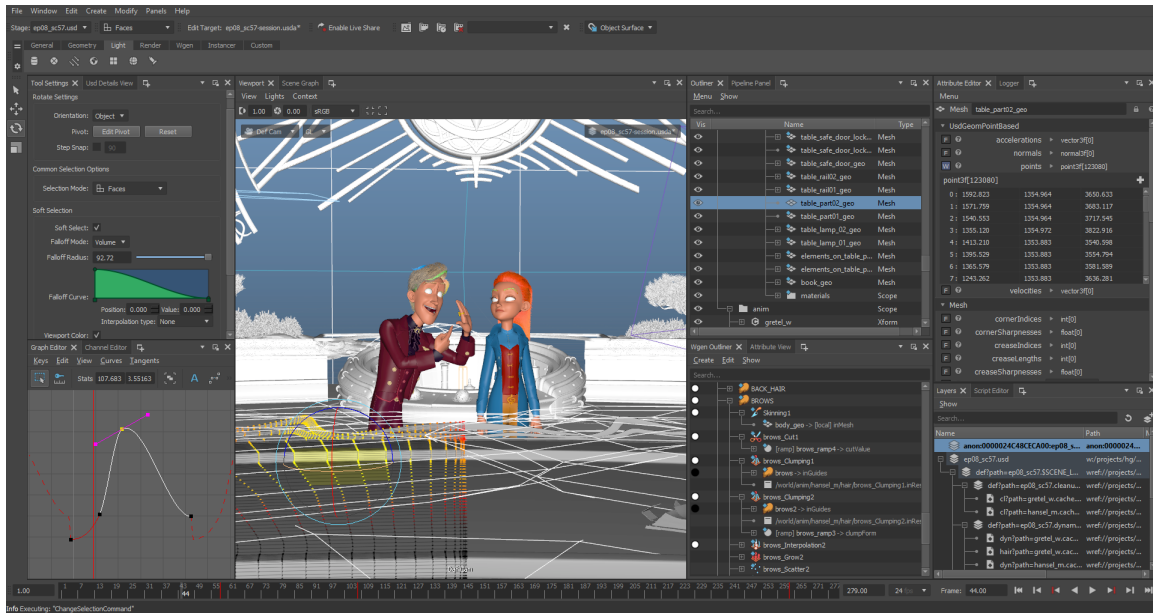


Figure 1: Wizart DCC Platform User Interface

ABSTRACT

As an indie animation studio for *Secret Magic Control Agency* (SMCA), we moved to Pixar's Universal Scene Description (USD) as the backbone of our pipeline. To make this possible, we introduced our in-house application framework, *Wizart DCC Platform* as a USD-native toolset. We used its extensibility to successfully implement new scene assembly, shading, hair grooming, and lighting workflows.

CCS CONCEPTS

• Computing methodologies → Graphics systems and interfaces.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '22 Talks, August 07-11, 2022, Vancouver, BC, Canada

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9371-3/22/08.

<https://doi.org/10.1145/3532836.3536259>

KEYWORDS

dcc, frameworks, pipeline, usd

ACM Reference Format:

Alexander Kalyuzhnyy, Konstantin Giliarovsky, Maksim Tokarev, Mikhail Kutsov, and Vlad Tagintsev. 2022. Wizart DCC Platform: extensible USD-based toolset. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks (SIGGRAPH '22 Talks)*, August 07-11, 2022. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3532836.3536259>

1 BACKGROUND

Our previous pipeline used our in-house scene description format to describe the assets to the third-party renderers, which worked well for us for years. Still, we continuously accumulated tech debt with our constrained resources.

The architecture of this system, was inspired by Cortex's ModularProcedural [Kaufman et al. 2013], including that we similarly built an abstract rendering interface that we used for the viewport OpenGL preview and final renderer translation.

To solve some of these problems, we designed the new procedural scene generation engine *SceneLib*. SceneLib could improve

lighting pipeline, but it could not help with our "direct" scene assembly workflows. We initially considered implementing an upgraded scene description format backed by the SceneLib engine but found that route could bring us again to the accumulation of tech debt, which we tried to avoid with our constrained resources. So instead, we looked into USD [Pixar Animation Studios 2016] as a possible solution that could solve our scene assembly issues.

Meanwhile, our inflexible hair pipeline needed to be upgraded as well. Our artists were frustrated with the current possibilities of the in-house hair system and the difficulties of pushing the curve data into the pipeline. To handle this better, we started to develop a new hair generation library *Wgen*.

USD gives us a compelling scene description system, and Hydra/Storm is a very modern and fast viewport technology. However, we found that most of the tools and DCC plugins that support USD editing or exports suffered from DCC/USD integration issues because they tried to marry two completely different scene descriptions. These tech issues burden R&D, TD, and artists since they had to be experts in USD, host DCC, and bridge DCC plugin implementation nuances. Instead, we thought about ditching host DCC interop problems altogether and writing our standalone USD editor, then all of these bridge issues could be solved much more easily.

2 USD EDITOR

The idea is not new. Standalone and native USD editing was explored before by Millefiori [Yudin and Ducatel 2019]. We developed our in-house extensible USD-native editor, Wizart DCC Platform, that we use as an application framework for all future tools:

- Comprehensive C++, Python API, plugin framework
- Extensible custom Hydra Engine, multiple Scene Delegates
- Viewport Tools API, rich manipulators, with basic modeling support
- AnimX [Autodesk Inc. 2017] toolset with Graph Editor for simple animation editing
- USDShade Node Editor for shading workflows
- Live share USD deltas using IPC between Wizart DCC and modeling package.

3 WGEN

We call our in-house node-based hair system, "Wizart Generator" (*Wgen* for short) because scattering and guide interpolation algorithms are highly inspired by Xgen [Thompson et al. 2003]. However, unlike Xgen, we decided to use OSL [Gritz et al. 2010] networks to modify the input parameters. We use the Wizart DCC Platform as a frontend to the *Wgen*. The *Wgen* data is represented in USD stage as custom *UsdWgen* schema, where we leveraged *UsdShade* for OSL networks, *USDGeom* standard prim types for mesh, and curve guides inputs. Artists manipulate USD prims directly where any change is tracked and can trigger recompute of the *Wgen* graph. To see computed curves in the viewport, on top *UsdImaging* *SceneDelegate*, we register a custom *WgenShape* *SceneDelegate*.

4 SCENELIB

Another toolset integrated on top of Wizart DCC is SceneLib Extension. SceneLib is a functional programming-inspired graph computation engine, idea is similar to Gaffer [Haddon et al. 2016], where

we execute compute graph on each location, but, unlike Gaffer, we use this computational architecture only for scenegraph processing.

It was used for all SMCA lighting/render workflows, directly translating data to the production renderer without Hydra because it was less mature when we developed this. We wrote the *USDIn* function for SceneLib to read scene data from the external USD files. Interestingly, our Hydra Engine has API to switch data sources and is not constrained to USD Stages. This way, lighting artists can preview SceneLib data in Viewport using SceneLib *SceneDelegate*. Similar to Animal Logic's Grip [Agland et al. 2020], the artist authors interconnected USD prims with our new *UsdSceneLib* schema. Each *UsdSceneLib* prim represented *SceneLibNode* which translated into one or more *Scenelib* functions, and function arguments based on *UsdSceneLibNode* attribute values.

5 CONCLUSIONS AND FUTURE WORK

All of this work made it possible to use Wizart DCC for various stages of production, including:

- Scene assembly of complex environments
- Shading of hero assets, environment, and props where the artists' author *USDShade* materials.
- Hero assets hair grooming using *Wgen* Extension
- Lighting and final rendering using SceneLib Extension

Wizart DCC Platform remains in active development, and we are working towards *open-sourcing* this technology.

ACKNOWLEDGMENTS

Thanks the whole Secret Magic Control Agency crew, who fearlessly embraced the new USD pipeline under tight deadlines and gave us valuable feedback. We would also like to thank Pixar's USD/Hydra teams made all of this work possible.

REFERENCES

- Steve Agland, Jakub Jeziorski, Manuel Macha, Simon Bunker, Francesco Sansoni, and Eoin Murphy. 2020. Grip and Filament: A USD-Based Lighting Workflow. In *ACM SIGGRAPH 2020 Talks* (Virtual Event, USA) (*SIGGRAPH '20*). Association for Computing Machinery, New York, NY, USA, Article 33, 2 pages. <https://doi.org/10.1145/3388767.3407350>
- Autodesk Inc. 2017. *AnimX Animation Library*. Retrieved May 1, 2022 from <https://github.com/Autodesk/animx>
- Larry Gritz, Clifford Stein, Chris Kulla, and Alejandro Conty. 2010. Open Shading Language. In *ACM SIGGRAPH 2010 Talks* (Los Angeles, California) (*SIGGRAPH '10*). Association for Computing Machinery, New York, NY, USA, Article 33, 1 pages. <https://doi.org/10.1145/1837026.1837070>
- John Haddon, Andrew Kaufman, David Minor, Daniel Dresser, Ivan Imanishi, and Paulo Nogueira. 2016. Gaffer: An Open-Source Application Framework for VFX. In *Proceedings of the 2016 Symposium on Digital Production* (Anaheim, California) (*DigiPro '16*). Association for Computing Machinery, New York, NY, USA, 31–42. <https://doi.org/10.1145/2947688.2947696>
- Andrew Kaufman, John Haddon, Ivan Imanishi, and Lucio Moser. 2013. Developing a Unified Pipeline with Character. In *Proceedings of the Symposium on Digital Production* (Anaheim, California) (*DigiPro '13*). Association for Computing Machinery, New York, NY, USA, 37–43. <https://doi.org/10.1145/2491832.2491837>
- Pixar Animation Studios. 2016. *Universal Scene Description Website*. Retrieved May 1, 2022 from <http://www.openusd.org>
- Thomas V. Thompson, Ernest J. Petti, and Chuck Tappan. 2003. XGen: Arbitrary Primitive Generator. In *ACM SIGGRAPH 2003 Sketches & Applications* (San Diego, California) (*SIGGRAPH '03*). Association for Computing Machinery, New York, NY, USA, 1. <https://doi.org/10.1145/965400.965411>
- Victor Yudin and Gregory Ducatel. 2019. Millefiori: A USD-Based Sequence Editor. In *Proceedings of the 2019 Digital Production Symposium* (Los Angeles, California) (*DigiPro '19*). Association for Computing Machinery, New York, NY, USA, Article 3, 4 pages. <https://doi.org/10.1145/3329715.3338882>